

LC 670: Maximum Swap

Company Tags: EXL, Google

■ Problem Understanding

Example 1:

num = 2736 → allowed to swap 2 numbers once to get highest number.

Swap 2 and 7 → t/b = 7236

Example 2:

9893 → here max num is 9, so swap 8, 9.

→ t/f = 9983

■ Approach & Logic

- For simplicity, first convert the num in array form.

Eg 2736 →

2	7	3	6
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- now try every i and j from $(i+1)$.
- for every i look for max value and its index to right.
- if this max value is $> arr[i]$ swap them, otherwise go to next i .

Eg here $arr[i] = 2$, max to right = 7 with idx...

∴ $(7 > 2)$ we swap.

■ C++ Code Implementation

```
class Solution {
public:
    int maximumSwap(int num) {
        vector<int> arr;
        queue<int> q;

        while (num > 0) {
            int dig = num % 10;
            arr.push_back(dig);
            num = num / 10;
        }

        // 2736 git converted to
        // [2,7,3,6]
        int n = arr.size();
        reverse(arr.begin(), arr.end());

        for (int i = 0; i < n - 1; i++) {
            int maxer = 0;
            int maxeridx = -1;

            for (int j = i + 1; j < n; j++) {
                if (arr[j] >= maxer) {
                    maxer = arr[j];
                    maxeridx = j;
                }
            }

            if (maxer > arr[i]) {
                swap(arr[i], arr[maxeridx]);
                break;
            }
        }

        int ans = 0;
        for (int i = 0; i < n; i++) {
            ans = ans + arr[i];
            ans = ans * 10;
        }

        return ans / 10;
    }
};
```